

A Practical Real-World Approach to Embedded Linux

The book definitely deserves shelf space in the library of anyone working in Embedded Linux.

In my humble opinion, *Embedded Linux Primer: A Practical Real-World Approach* is perhaps not the best place a newbie would want to get started, since it lacks a hands-on style to holistic embedded Linux development. The hands-on style is, however, found in places where the author has talked about BusyBox, some of the profiling and debugging tools, and the like. For anyone moderately familiar with the Linux kernel sources and concepts, a number of chapters should be easy browsing.

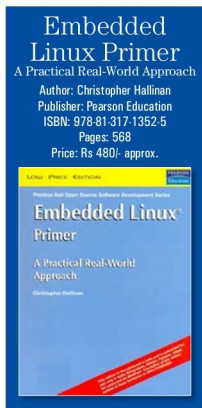
The book starts off with an introductory chapter and talks a bit about the state of embedded Linux, its soaring popularity and the reasons behind it.

The next chapter, titled 'Your First Embedded Experience', familiarises the reader with the basic anatomy of an embedded Linux system—the initialisation process, the storage options available, cross-development, and concludes by taking a look at some embedded Linux distributions.

Chapter 3 talks about 'Processor Basics'. As the name suggests, it familiarises the reader with the various processor architectures—from standalone processors and integrated processors, to the hardware platforms used in embedded systems.

The chapter titled 'The Linux Kernel—A Different Perspective', introduces the compilation of the kernel sources—the Linux kernel build system and configuration options are introduced here. There's nothing new for someone already familiar with how to compile the Linux kernel.

Chapter 5 (Kernel Initialisation) and Chapter 6 (System Initialisation),



together show how the Linux kernel and filesystems fit into the picture and give you a working environment. These chapters talk in some detail about the *init* process, initial RAM disk, and the Linux filesystems—Filesystem Hierarchy Standard (FHS), root filesystem, minimal filesystems, etc.

Chapter 7 is dedicated to bootloaders and their role in an embedded system. Together with Chapters 4 and 5, the reader gets a clear idea of how the various pieces fit in a working embedded Linux system. Chapter 8 talks about Linux device drivers.

The next one is dedicated to the popular filesystems being used in embedded Linux systems today. This chapter also covers JFS2, which is an

important embedded filesystem. The reader is also introduced to the murky area of building a filesystem image from scratch, when what's available does not serve the purpose.

The chapter on MTD subsystems covers Memory Technology Devices, which is an abstraction layer between the low-level device complexities, and the higher-layer data organisation and storage formats that use memory devices.

Chapter 11 introduces the Swiss-army knife of embedded Linux development—BusyBox. This chapter, along with Chapters 12 and 13, talks about setting up an embedded development environment—with cross-compilers, profiling, tracing and debugging tools.

Chapter 14 is about kernel debugging techniques, and following it are debugging techniques for user-space application programs in Chapter 15.

In the next chapter, 'Porting Linux', the author takes a look at the various issues involved in porting Linux to a custom board. The book finally ends with real-time Linux in Chapter 17.

As I mentioned earlier, I would place this book in the reference section of my bookshelf, and pick it up whenever I am faced with one of the issues it deals with. The book would be of great help to embedded Linux developers who have some work experience. **END** 

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